

Davide Avagnano

Full-Stack Software Engineer — TypeScript · Node.js · PostgreSQL · Applied AI (LLM / RAG)

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Profile

Full-stack engineer who ships **production systems end-to-end**. Designed and built a multi-tenant SaaS **almost single-handedly** — ~57k LOC, 31 tables, ~145 type-safe procedures — from architecture to deployment, and contributed to high-volume marketing-intelligence platforms (160k+ LOC, 15+ distributed workers). Center of gravity on **backend and architecture**: data modeling, type-safe APIs, async workers, security. Runs **applied AI in production** — RAG, streaming assistants — with usage logs and spend caps, not as a gimmick. Engineer by training (BSc Aerospace), self-taught developer by choice.

Experience

Software Engineer — Freelance / Independent Consultant

Feb 2025 – Present

Full-time consulting across multiple clients: product SaaS, high-volume data platforms, SecOps tooling.

Multi-tenant booking SaaS — end-to-end product ownership

2026 – Present

- Designed and built **almost single-handedly** a multi-tenant SaaS (**Next.js 16 App Router/RSC, React 19, tRPC, Drizzle ORM, PostgreSQL, Redis**): **~57k LOC** TypeScript, 31 tables, ~145 type-safe procedures, 6 scheduled workers — from architecture to production deploy.
- Built an **applied AI layer with cost governance**: streaming conversational assistant (Vercel AI SDK) and automated translation across 5 languages, with usage logs and **per-model spend caps**.
- Granular two-axis authorization (RBAC scopes + plan gating) and **PII at-rest encryption** (AES-256-GCM) with boundaries enforced in CI; multi-channel notifications (email + push) via outbox pattern and idempotent workers (BullMQ/PM2).
- Living documentation** (50+ technical and functional docs) and planned refactoring as part of "done"; deployed on Linux (Nginx, CI/CD via GitHub Actions).

Marketing-intelligence & e-commerce analytics platforms

2025 – 2026

- Developed and analysed **complex production code** on high-volume platforms (Node.js/TypeScript, PostgreSQL, Redis, Elasticsearch): **config-driven ETL**, cross-platform data reconciliation and distributed workers (15+ PM2 processes).
- Built **20+ external API integrations** (Meta Graph, Shopify GraphQL, TikTok, Taboola, GA4, call tracking) through reusable connectors.
- Shipped **AI features in production**: RAG assistant with **hybrid BM25 + kNN retrieval** over Elasticsearch, dynamic SQL generation, and LLM content moderation.
- Delivered analytics dashboards (React, Recharts) with multi-level RBAC, and a **SecOps automation tool** (TypeScript + Bash): Suricata IDS + Elasticsearch + UFW, remote rule lifecycle via SSH, ELK stack.
- Worked in **small teams** (3–5 developers) alongside CTO, design and marketing; ran requirements-gathering calls and supported colleagues on technical problems.

Skills

Languages: TypeScript (advanced), JavaScript, SQL, Bash; Python (data analysis, scripting)

Backend & architecture: Node.js, Express, tRPC (type-safe APIs), Zod, REST; event-driven design and distributed workers (BullMQ, PostgreSQL LISTEN/NOTIFY); domain modeling; multi-tenant and config-driven architectures

AI / LLM: OpenAI API, Vercel AI SDK (streaming), **RAG** (hybrid BM25 + kNN), conversational assistants, content moderation, LLM **cost governance** (usage logs, per-model spend caps)

Data & persistence: PostgreSQL (advanced), MySQL, Redis, Elasticsearch, Drizzle ORM; config-driven ETL and cross-platform data reconciliation; pandas, NumPy

Frontend: React (17→19), Next.js (App Router / RSC), Tailwind CSS, shadcn/ui, TanStack Query, Zustand, i18n

Security: authentication (better-auth / NextAuth, OAuth, JWT), granular RBAC, PII encryption (AES-256-GCM), rate limiting, GDPR/consent; SecOps (Suricata IDS, ELK, UFW, SSH)

DevOps & UNIX: Linux (Ubuntu), Docker, Nginx, PM2, CI/CD (GitHub Actions), Git; production deploys on VPS

Education & Background

BSc in Aerospace Engineering — University of Naples Federico II

2016 – 2020 · 105/110

Career transition to software — self-taught: Python (data analysis, algorithms), then web and full-stack (React/Next.js/Node.js); personal projects and study. 2021 – 2024

Languages

Italian (native) · English **B2** — daily professional use: technical documentation, English-language codebases and multilingual products.